SEAT, WATER-CLOSET (LAND USE)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for water closet seats with and without cover.

1.1.1 Specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available. This specification is intended to cover water closet seats generally used by the Federal Government.

1.2 Classification.

1.2.1 Type/specification part number (SPN). Water closet seats covered by this specification shall be of the type indicated by the SPN (see table I).

TABLE I. SPN identification

SPN	Shape	Cover included	Color	Ref. par.
QRTW	Regular	Yes	White	3.12.a.
CRWX	Regular	No	White	3.12.b.
CEBX	Elongated	No	Black	3.12.c.
CEWX	Elongated	No	White	3.12.d.

1.2.2 Definitive part number (DPN). Water closet seats covered by this specification shall be identified by a DPN. The DPN shall consist of the Federal Specification symbol followed by a hyphen and the SPN. This part number is intended for cataloging and ordering purposes; it is not intended for surface printing on seats. The DPN shall be written as indicated below (see 6.2):

DPN---------- WWS1912 - XXXX

Fed. Spec. symbol
SPN (see 1.2.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

B 117 - Method of Salt Spray (Fog) Testing.
B 380 - Corrosion Testing of Decorative Chromium Plating by the Corrodkote Procedure.
D 2444 - Test for Impact Resistance of Thermoplastic Pipe and Fittings by Means of a Tup (Falling Weight).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Hinge assembly. Unless otherwise specified herein, the hinge assembly consisting of the hinge, post, and fastener shall be of any of the following materials at the manufacturer's option:

(a) Chromium plated copper alloy.
(b) Corrosion-resisting steel, AISI 200 or 300 series, or 430F.
(c) Plastic covered, copper alloy.
(d) Plastic covered, corrosion-resisting steel, AISI 200 or 300 series, or 430F.
(e) Plastic covered, zinc-coated steel, or coated steel conforming to 3.10.2.
(f) Plastic covered, anodized aluminum alloy.
(g) Molded or fiber reinforced, (PA), acetal, or any comparable plastic for plastic seats.

3.1.2 Plastics. Plastics shall include acetal, acrylic, fluorcarbon, nylon, polypropylene, acrylonitrile butadiene-styrene, polystyrene or any other suitable plastic including fiber reinforced plastics.

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3.1.3 Rubber. Rubber shall be natural or synthetic.

3.2 Construction.

3.2.1 Seats and seat covers. Seats and seat covers shall be solid molded plastic.

3.2.2 Hinges, posts (studs), and fasteners.

3.2.2.1 Hinges. Hinges shall be free rotation type, check type, or self sustaining as specified by the specification part number.

a. Free rotation type. Free rotation type hinges shall permit seats and covers to be rotated freely by hand.

b. Check hinge type. Check hinges shall restrict rotation of seats within an 11 degree angle back as measured from the vertical position.

c. Self sustaining type. Hinges shall be adjustable and shall maintain the seat in any position by friction. Hinges shall permit the seat to swing easily, but not drop or slam against the bowl or back stop. The mechanism for setting or resetting the degree of friction shall not require the use of special tools and shall not require removal of the seat from the bowl.

3.2.2.2 Hinge posts and pintles. Hinge posts and pintles shall be of metal or plastic. Metal posts for plastic encased hinges shall be coated or encased in plastic above the water closet rim. Plastic posts shall have a major thread diameter of not less than 0.375 inch. The diameter of plastic pintles shall not be less than 5/16-inch. The diameter of metal posts and pintles shall be in accordance with acceptable industry practice. All threads shall be in accordance with acceptable industry practice. The color of plastic coated or encased hinges and posts shall be the same as the seat, or the seat and cover.

3.2.2.3 Seat hinge fasteners and washers. Seat hinge fasteners and washers shall conform to normal industry practice for the type seat furnished.

3.2.3 Bumpers. Bumpers shall be of rubber, rubber-like, or any suitable plastic material. The bumper shall have an appearance comparable to the seat. Two or more bumpers shall be securely attached to the underside of each seat and of each cover. Seats shall be 1/8 to 1/2 inch above the rim of the bowl. Covers shall be 1/8 to 1/2 inch above the seat. Bumpers on seats and covers shall be elongated or round. Method used for attachment of bumpers shall be at the manufacturer's option.

3.3 Surface configuration. Seats shall have a flat bottom or a concave bottom. The upper surface of seats shall be rounded and suitably shaped. Covers shall be flat or concave, and shall conform to the pattern of the seat.

3.4 Dimensional requirements.

3.4.1 Thickness. Minimum thickness of seats midway between the front and the centerline of rotation in the back, and the minimum thickness of covers at or near the geometric center shall be in accordance with table II. The thickness shall be measured as specified in 4.4.1.

TABLE II. Thickness of seats and covers, minimum (inches)

Type	Seat	Cover
QRTW	5/8	1/4
CRWX	3/4	-
CEBX	3/4	-
CEWX	3/4	-

3.4.2 Seat dimensions. The seat dimensions indicated in figure 1 shall be as specified in table III.

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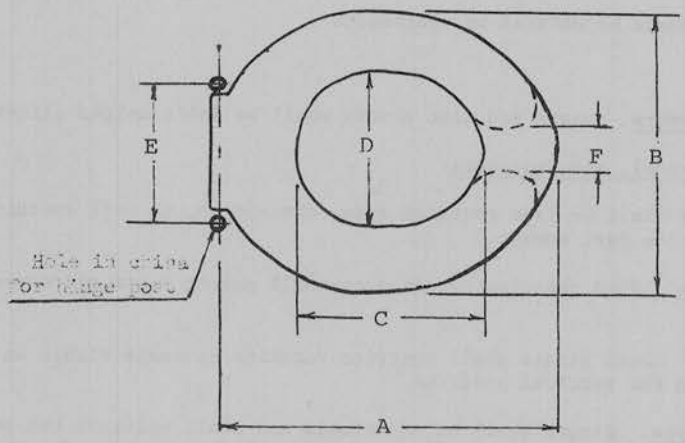


FIGURE 1. Seat dimensions

TABLE III. Seat dimensions

Dimension	A	B	C	D	E 1/	F
Regular	16 to 17	14-1/4 min	9 min	7-3/4 min	5-1/2	3
Elongated	17 to 18	14-1/4 min	9 min	7-3/4 min	5-1/2	3

1/ The "E" dimension is a specific measurement; it is not a minimum dimension.

3.5 Color. Seats and covers shall be black or white in accordance with the specification part number. Seats and covers shall be the same color throughout.

3.6 Impact resistance. All seats and covers, when tested in accordance 4.4.2 shall not chip, crack, or break and shall not exhibit any noticable effect other than a slight indentation mark at the point of impact.

3.7 Flexural strength and load capability. Seats and covers shall not deflect more than 0.62 inch; the permanent set shall not exceed 0.10 inch, nor shall they fracture when tested in accordance with 4.4.3.

3.8 Dimensional stability and moisture resistance. Seats and covers shall not distort and dimensions shall not increase in thickness due to moisture absorption (see 4.4.4).

3.9 Staining and chemical attack. When tested in accordance with 4.4.5 there shall be no evidence of chemical attack or change in surface appearance of all seats and covers.

3.10 Metal finishes.

3.10.1 Chromium plated and zinc-coated metal. Chromium plated and zinc-coated metal shall not exhibit any products of corrosion when tested in accordance with 4.4.6.

3.10.2 In lieu of the zinc-coated steel specified in 3.1.1 (e), any other type of inorganic corrosion-resisting coating on steel that will not exhibit products of corrosion when tested in accordance with 4.4.6 may be used. Coatings in accordance with this paragraph shall equal or exceed all characteristics of zinc-coated steel.

3.11 Marking. Seats shall be marked with the name of the manufacturer or the manufacturer's trademark. Markings shall be permanent.

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3.12 SPN identifier. The four (4) position SPN identifier code shall specify the water closet seat as below. All dimensions shall be as specified in 3.4.

SPN

a. QRTW. White plastic regular seat, closed front, white plastic top, extended back, standard hinge with plastic surface treatment, residential seat.

b. CRWX. White plastic regular seat, open front, without top cover, extended back, standard hinge with plastic surface treatment, industrial seat.

c. CEBX. Black plastic elongated seat, open front, without top cover, extended back, seat check hinge with plastic surface treatment, industrial seat.

d. CEWX. White plastic elongated seat, open front, without top cover, extended back, self-sustaining hinge with plastic surface treatment, industrial seat.

3.13 Workmanship. Water closets and all parts shall be free from any defects which affect serviceability or operation. Metal and plastic parts shall be free from cracks. Welds in fabricated fixture supports shall be free from cracks and closely spaced in line surface porosity. Plastics shall be free from mold defects which impair servicability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. A lot shall consist of all water closet seats with the same DPN offered for delivery at one time.

4.2.2 Sampling for examination and tests. Select sample size for examination in accordance with level S-4 in MIL-STD-105. Select sample size for tests in accordance with level S-2 in MIL-STD-105.

4.3 Examination and tests.

4.3.1 Examination. Examine water closet seats for defects listed in table IV. The acceptable quality levels (AQL's) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

TABLE IV. Classification of defects, end item

Defects	Major	Minor
Item not in accordance with specification part number specified, (see 1.2.1)	X	
Hinges, hinge posts or fasteners not as specified, (see 3.2.2).	X	
Bumpers not as specified, (see 3.2.3)		X
Dimensions not in accordance with 3.4.2	X	
Color not as specified (see 3.5)		X
Marking not as specified (see 3.11)		X
Damage or defects affecting function or serviceability (see 3.13)	X	
Damage or defects not affecting function or serviceability (see 3.13)		X

4.3.2 Test AQL. The AQL for all tests shall be 4.0 percent defective.

4.4 Test methods.

4.4.1 Seat thickness measurement. Use any suitable method. The measurement shall determine the thickness of the seat or cover material for conformance to 3.4.1.

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4.4.2 Impact. Test for impact using tup B in accordance with ASTM D 2444.

4.4.2.1 Specimen holder.

a. Seats.
(1) Remove bumpers.
(2) Position seat in holder so that seat rests on supports perpendicular to the drop of the tup.
(3) Place one support under front of seat along a line of the removed bumpers.
(4) Place a second support in a similar position to the rear of the seat.
(5) The free span between supports shall be 8 inches for regular seats and 10 inches for elongated seats.
(6) Clamp seat firmly in position.

b. Covers. Place covers in holder using the procedure stated in 4.4.2.1, a.

4.4.2.2 Test. Use a tup weighing 8 pounds for plastic seats and covers. Adjust height of fall to 15 plus or minus 1/8 inches. Position covers so that tup impacts on the crown of the cover. Position seats so that the tup impacts on the top side at two diameterically opposite locations (see 3.6).

4.4.3 Load tests. Test at an ambient temperature of 70° ± 5°F. (see 3.7).

4.4.3.1 Open front seats. Position the seat in a vertical plane by placing the hinge end on a temporary support and suspending the front end of the seat at a point 1 inch in from the front on the edge of the front opening. The seat shall be suspended in such a manner that the points of support and suspension are on a horizontal or near horizontal plane. From a point opposite the point of suspension, gradually apply a downward load of 250 pounds. Edges of seat may be slightly notched in way of loading. Retain load for one minute and release. Examine for conformance to 3.7.

4.4.3.2 Closed front seats. Apply a 300 pound load to plastic seats. Include weight of channels and connecting plate assembly used for static testing water closets.

4.4.4 Dimensional stability and moisture resistance. Place seats and covers on a flat surface and record the amount of warpage determined with feeler gages. Measure and record thickness of seats and covers. Determine thickness using a micrometer at 4 separate locations separated by at least 3 inches. Clean seats and covers using a suitable solvent and place them in an environmental chamber maintained at 140° ± 4°F and a relative humidity of 95 ± 5 percent for a period of at least 24 hours. Remove seats and covers from chamber and wipe dry. Repeat measurements for thickness and warpage (see 3.8).

4.4.5 Chemical attack. Immerse a portion of seats and covers in the following solutions maintained at a temperature of 77° ± 2°F:

Chemical solutions, and water phenol solution, 0.5 percent concentration.	Time, hours 1/2
Trisodium phosphate solution, 1.0. percent.	24
Distilled water.	24

Prepare the two solutions as follows:

(a) Phenol solution. Dissolve 47g of carbolic acid crystals USP in 950 ml of water.
(b) Trisodium phosphate solution. Dissolve 21g of $Na_3PO_4 10H_2O$ or 23g of $Na_3PO_4 12H_2O$ crystals USP in 100° ml of water.

Tests may be run concurrently using several seats and covers. Remove seats and covers from the solutions and clean with distilled water. Dry seats and covers and examine exposed areas for staining and chemical attack (see 3.9).

4.4.6 Test plated metals and plastics for corrosion resistance in accordance with ASTM B 380 using a single test period of 20 hours ± 15 minutes or a salt spray test for 24 hour period in accordance with ASTM B 117 (see 3.10).

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TABLE V. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted: incorrect; illegible; improper size, location, sequence or method of application.
Material	Any compound missing, damaged or not conforming to the applicable specification or requirements stated herein.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Quantity per container is more or less than required. Gross weight exceeds requirements of the box specification.

5. PREPARATION FOR DELIVERY.

5.1 Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. Each seat shall be packaged in a close-fitting box conforming to PPP-B-636, class weather resistant, style FOL. Close and waterproof seal the boxes in accordance with the appendix to PPP-B-636.

5.1.2 Level B. Each seat shall be packaged in a close-fitting box conforming to PPP-B-636, class domestic, style FOL. The boxes shall be closed in accordance with the appendix to PPP-B-636.

5.1.3 Level C. Each seat shall be packaged to insure protection during shipment and safe delivery to destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Six seats packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. Close, waterproof seal and strap the boxes in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Six seats packaged as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-636, class-domestic. Closure of boxes shall be in accordance with Method II of the appendix to PPP-B-636.

5.2.3 Level C. Seats packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

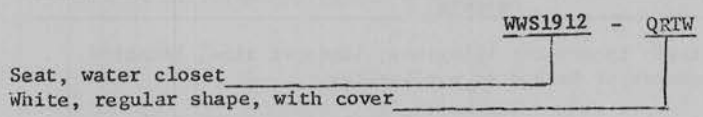
6.1 Intended use. Water closet seats covered by this specification are intended for residential and regular industrial use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Specification part number (see 1.2.1).
(c) Selection of applicable packaging and packing levels (see 5.1 and 5.2).
(d) Whether marking shall be in accordance with 5.3.1 or 5.3.2 (see 5.3).

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6.3 Example of specification part number.



Preparing activity:

GSA-FSS

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